

Retail Growth & Inventory Optimization Strategy for Maven Toys

Maven Toys is a multi-city toy retail chain operating across Mexico. This project combines **Retail Analytics, Inventory Analytics, and Supply Chain Analytics** – using PostgreSQL and Power BI – to identify operational inefficiencies and deliver data-driven recommendations for inventory optimization and business growth.



Business Objectives & Dataset



Key Objectives

- Analyze sales performance across products, stores, and cities
- Identify stockout risks and overstocked inventory segments
- Build interactive executive dashboards for decision support

Dataset (~800K transactions)

- **Sales:** Transaction records with units sold
- **Products:** Categories, cost, and pricing
- **Stores:** 50 stores across 29 cities
- **Inventory:** Stock on hand by store and product

Revenue & Sales Analysis

\$14.44M

Total Revenue

\$4.01M

Total Profit

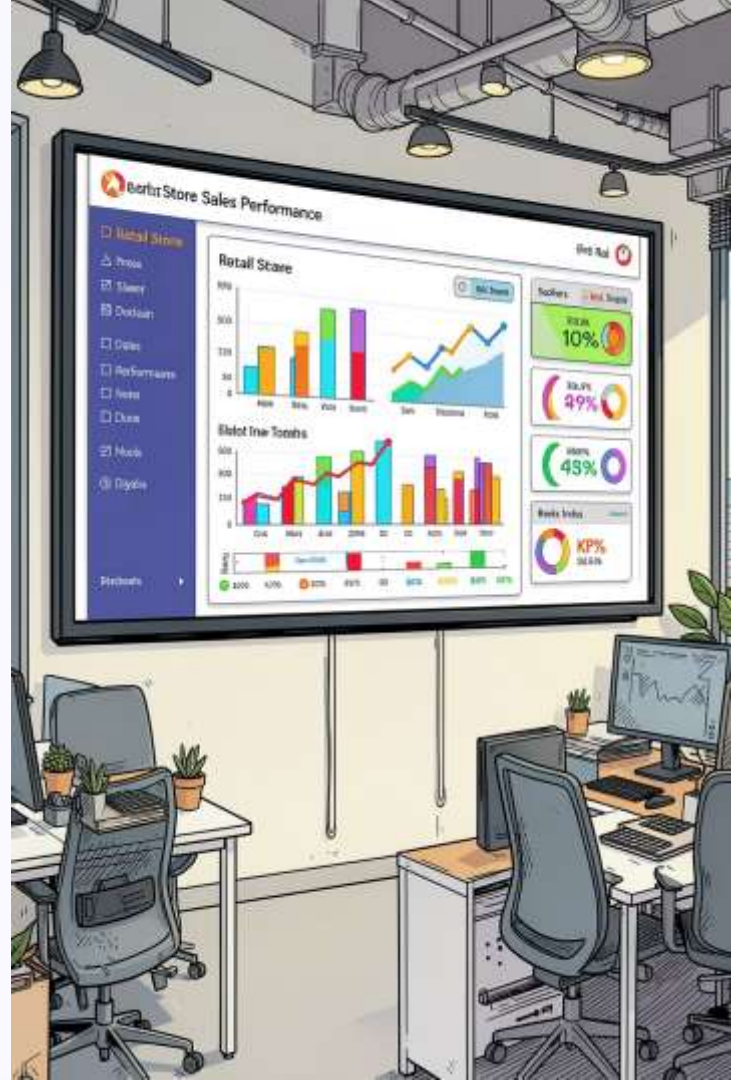
1.09M

Units Sold

27.79%

Avg Profit Margin

- **7 cities (25%)** generate 50% of total revenue
- **Toys** contribute ~35% of revenue; **Games & Electronics** show strong margins (30% & 44%)
- **Q4 holiday** drives significant revenue spikes; Toys & Sports-Outdoors peak March - June
- **Electronics** revenue declining; **Art & Crafts** showing consistent growing demand



Store Performance Analysis



Key Findings

- **Airport stores** achieved the highest average revenue per store and stronger inventory turnover
- **Downtown stores** consistently outperformed Residential locations in revenue and profitability
- Revenue spread between top and bottom stores is 4x, but largely uniform across most locations
- Several underperforming stores maintained high inventory despite lower sales activity
- No evidence linking store age to revenue performance

30K

Inventory Units

29

Cities

50

Stores

Product, Profitability & Inventory Analysis

Product KPIs

- **35 products** total; top seller: **Lego Bricks**
- Avg selling price: **\$13.25**
- ~40% of products drive **80% of revenue**

Product Segmentation

- **High Rev / High Profit:** Colorbuds, Action Figure
- **High Rev / Low Profit:** Magic Sand, Rubik's Cube
- **Low Rev / High Profit:** Mini Basketball Hoop, Plush Pony
- **Low Rev / Low Profit:** Classic Dominoes, Playdoh Toolkit

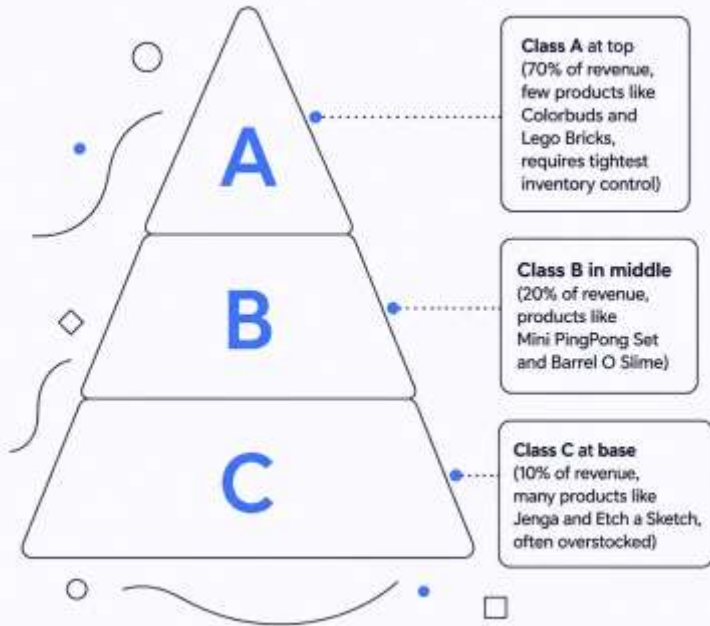
Inventory KPIs

- Total Inventory Value: **300K units**
- Stock Coverage: **17 days**
- Inventory Turnover: **36.67**

Key Findings

- **Action Figures** at critically low stock; **Dinosaur Figures** significantly overstocked
- Inventory allocation uneven across stores and regions
- Electronics: ~45% margin but consistent revenue decline

ABC Classification & Pareto Analysis



ABC Classification

- **Class A:** ~70% of revenue – tightest inventory control needed (e.g., Colorbuds, Lego Bricks)
- **Class B:** ~20% of revenue – moderate focus (e.g., Mini PingPong Set)
- **Class C:** ~10% of revenue – often overstocked (e.g., Jenga, Etch a Sketch)

Pareto Insight

~40% of products generate nearly **80% of total revenue**. Revenue concentration is highly uneven – strategic focus on high-performing products can significantly improve profitability and inventory efficiency.

Key Business Insights

Revenue Insights

- Revenue growth concentrated in a small number of product categories
- Downtown stores consistently generated superior performance metrics

Inventory Insights

- Several high-demand products faced elevated stockout risk
- Overstock conditions existed within low-performing product categories
- Inventory allocation did not consistently align with historical demand patterns

Operational Insights

- Inventory turnover efficiency differed significantly across product categories
- Product profitability and sales volume were not always aligned



Strategic Recommendations 1-3



Increase High-Demand Inventory

Boost allocation for high-demand, high-turnover products with elevated stockout risk.

Impact: Reduced lost sales, improved customer satisfaction, increased revenue capture



Reduce Overstocked Inventory

Cut exposure for low-turnover and Class C products to lower holding costs.

Impact: Lower holding costs, improved inventory efficiency, better working capital



Regional Inventory Strategies

Adopt city-specific allocation based on localized demand patterns.

Impact: Improved inventory accuracy, reduced regional imbalances, enhanced responsiveness

Strategic Recommendations 4-6

1

Prioritize Class A Products

Focus replenishment, marketing, and shelf allocation on top revenue contributors.

Impact: Higher revenue growth, better inventory utilization, increased operational focus

2

Improve Demand Forecasting

Leverage historical sales trends and seasonality patterns for future inventory planning.

Impact: Better procurement decisions, reduced stockout frequency, improved planning accuracy

3

Optimize Store-Level Inventory

Rebalance allocation based on store demand, sales velocity, and regional performance.

Impact: Lower excess inventory, improved stock availability, enhanced operational efficiency

Conclusion

This project combined **Retail, Inventory, and Supply Chain Analytics** to deliver actionable business intelligence for Maven Toys – identifying major opportunities for revenue optimization, inventory efficiency, and strategic allocation.

Revenue Optimization

Inventory Efficiency

Operational Enhancement

Strategic Allocation

- ✓ By integrating SQL analytics, Power BI visualization, and business strategy recommendations, this project provides a **scalable framework** for retail performance optimization and operational decision-making.

